INCH-POUND

MIL-C-17361/3(SH)
10 March 1989

MILITARY SPECIFICATION SHEET

CIRCUIT BREAKERS, AIR, ELECTRIC, INSULATED HOUSING (SHIPBOARD USE) TYPE AQB - LF100 (FOR REPLACEMENT ONLY)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-17361.

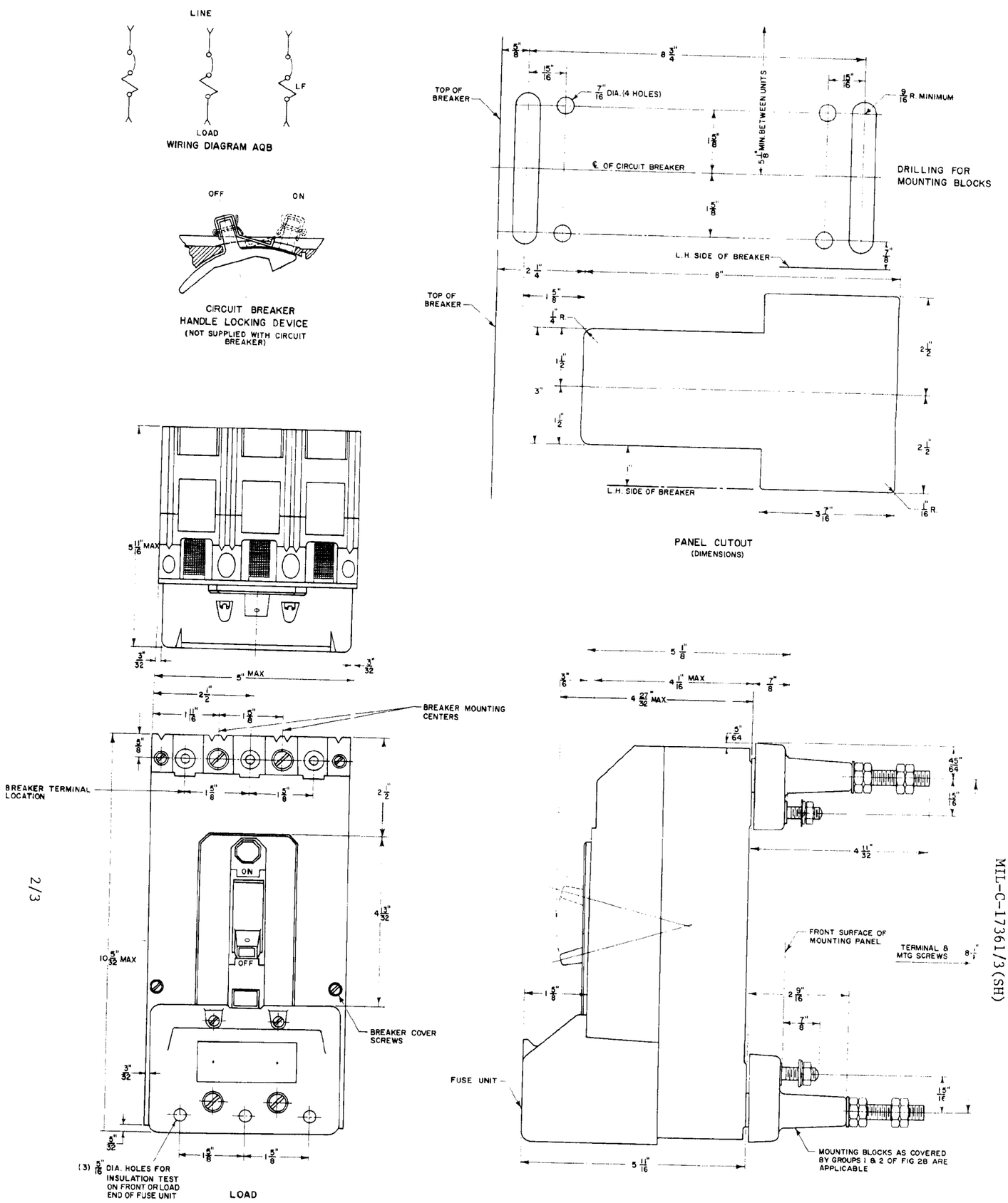
AMSC N/A FSC 5925

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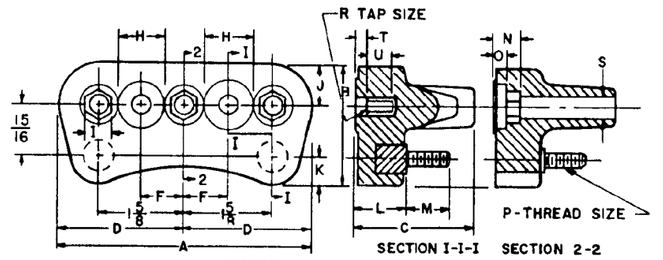
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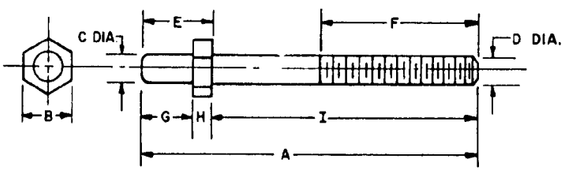


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FIGURE 1. Circuit breaker type AQB-LF100, outline and mounting dimensions.

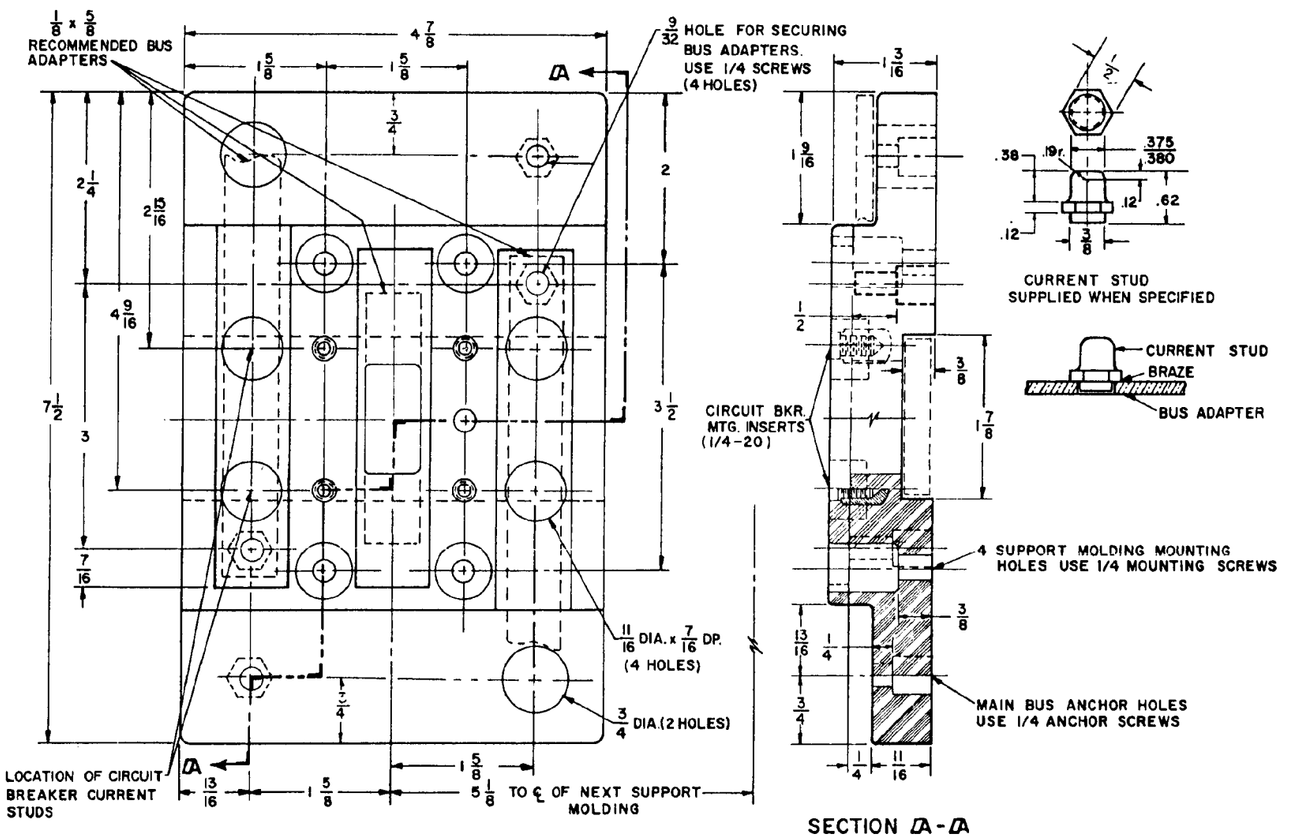


A	B	C	D	F	H	I	J	K	L	M	N	O	P	R	S	T	U
4 21/32	2 9/64	2 9/16	2 21/64	13/16	7/8	.515	45/64	1/2	7/8	7/8	17/32	13/32	5/16-18	1/4-20	25/64	3/16	1/2



A	B	C	D	E	F	G	H	I	THREAD SIZE
4 3/4	1/2	3/8	3/8	15/16	2	11/16	1/4	3 13/16	3/8-16

GROUP 1
DETAILS OF TERMINAL MOUNTING BLOCK & TERMINAL STUD



GROUP 2 DETAILS OF MTG. BLOCK FOR LINE END OF TWO AQB-A100 CIRCUIT BREAKERS MOUNTED IN DISTRIBUTION PANEL.

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FIGURE 2. Mounting blocks for circuit breaker type AQB-LF100, outline and mounting dimensions.

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REQUIREMENTS:

(NOTE: Type AQB - LF100 circuit breakers shall be used for replacement only.)

NQB circuit breakers shall consist of an AQB type without trip unit and a modification kit of metallic parts.

Dimensions: See figures 1 and 2.
Rated continuous current: 100 amperes.
Voltage and frequency: 500 Vat, 60 Hz.
355 Vdc
Interrupting rating: 100,000 amperes, 60 Hz.
150,000 amperes, 250 Vdc.
150,000 amperes, 355 Vdc.
Peak let-through current: LF5H100RE, 20,000 amperes.
LF5H100RD, 35,000 amperes.
Trip unit: Interchangeable.
Trip unit rating (amperes): See table I.
Protection: LTD, instantaneous, and fuse - all 3 poles.
LTD calibration: See table 11.
Instantaneous calibration: See table I.
Current limiting fuse: LF5H100RE, 5,000 - 15,000 amperes.
LF5H100RD, 10,000 - 15,000 amperes.
Type of connections: Back (removable mtg.).
Electric operator: Not available.
Undervoltage trip or shunt trip: None.
Auxiliary switch: None.
Handle lock: Group 2. (1 of MIL-C-17361)
Handle boot: Not available.
Electrical test: The rate of operation shall be 6 operations per minute for a total of 6000 close-open operations.
Temperature rise tests: See table III.
Interrupting tests: See table IV.

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TABLE I. Instantaneous trip calibrations.

Fuse Style	Trip element rating (amps) 1/	Instantaneous trip settings - amperes/maximum time 2/	
		Current (amperes)	Time (seconds)
LF5H100RE	15D	195	0.024
	25D	325	
	25DF	195	
LF5H100RD	50D	650	0.024
	50DF	325	
	75D	975	
	75DF	487	
	100D	1300	
	100DF	650	
	100DG	2000	

1/ Identification marking of trip units:
Numerals indicate continuous current rating of the specific trip unit.
Letter "D" denotes that the trip units are applicable to circuit breaker type AQB-LF100.
Letter "F" denotes calibration for applications having low inrush currents.
Letter "G" denotes special calibration for use as generator or transformer protection.

2/ The tolerance for instantaneous trip shall not exceed plus or minus 10 percent. Time shall be measured at three times pickup current. For qualification tests only, not for routine calibration.

TABLE II. Tripping characteristics - long time delay.

LTD trip element percent current	Tripping time 60 Hz ac trip element
130	Not less than 1 hour
225	Less than 1 hour
600	Between 20 and 32 seconds

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TABLE III. Size of cable and bus bars for temperature rise tests.

Copper rating-amperes	Cable size	Copper bus size (inch)
100	AWG 3	3/4 x 1/8

TABLE IV. Interrupting test.

Circuit sample	Element amperes	Interrupting amperes	Power factor lagging	Unit operation	See note
1A	100	100,000	0.15-0.20	O-CO-CO	2/
1A(1)	100	10,000	0.45-0.50	O-CO-CO	3/ 4/
1A(1)		15,000	0.25-0.30	O-CO-CO	2/ 3/
1A(1)		25,000	0.15-0.20	O-CO-CO	2/ 3/
1A(1)		50,000	0.15-0.20	O-CO-CO	2/ 3/
lB	15	100,000	0.15-0.20	O-CO-CO	2/ 3/
1B(l)	25	5,000	0.45-0.50	O-CO-CO	3/ 4/
1B(l)		15,000	0.25-0.30	O-CO-CO	2/ 3/
1B(l)		25,000	0.15-0.20	O-CO-CO	2/ 3/
1B(l)		50,000	0.15-0.20	O-CO-CO	2/ 3/
1B(2)	50	100,000	0.15-0.20	O-CO-CO	2/ 3/

Oscillograms shall be taken on all interrupting tests.
Circuit breaker shall open and at least one fuse blow.
New circuit breakers may be used for tests above the standard circuit breaker rating.
Circuit breaker shall open on instantaneous trip and fuses shall not blow.

Preparing activity:
Navy - SH
(Project 5925-N103-03)

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